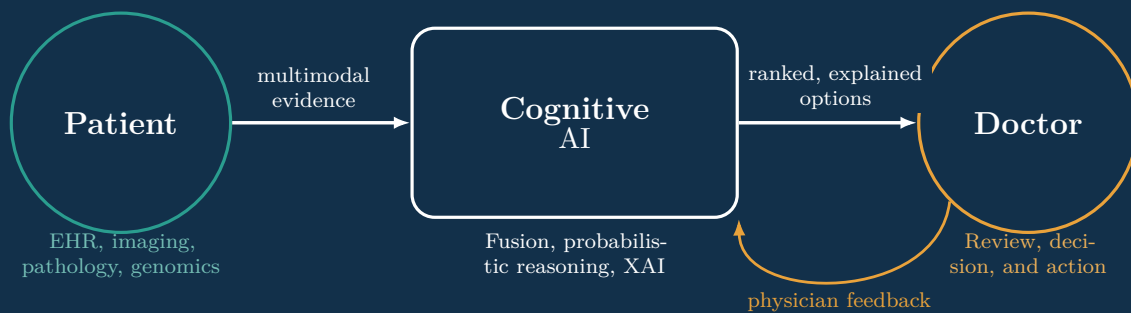


AI-Assisted Cancer Diagnosis

Human Judgment and Cognitive Systems

A multimodal, explainable clinical decision-support architecture for resolving diagnostic uncertainty while preserving physician authority.



Scenario. Three oncologists propose Stage II lung cancer, tuberculosis, and a benign pulmonary nodule for the same patient. The design below converts disagreement into an auditable differential diagnosis and a safe plan for confirmatory testing.

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1 Design Position and Clinical Goal

The proposed platform is a **clinical decision-support system (CDSS)**. It augments oncologists by organizing evidence, quantifying uncertainty, retrieving relevant knowledge, and explaining alternatives. It does not replace a physician, independently declare a final diagnosis, or silently retrain itself in production.

Core objective: transform heterogeneous evidence into a ranked differential diagnosis, show why each alternative is plausible or weak, identify the next test with the greatest expected diagnostic value, and preserve a complete human-review trail.

1.1 Design principles

- **Patient-specific:** combine longitudinal context, not isolated measurements.
- **Multimodal:** jointly interpret text, images, laboratory values, genomics, and time series.
- **Probabilistic:** represent ambiguity instead of forcing a binary answer.
- **Explainable:** expose supporting, opposing, and missing evidence.
- **Human-governed:** require clinician confirmation for consequential actions.
- **Continuously improved, not continuously unstable:** deploy only validated model versions.

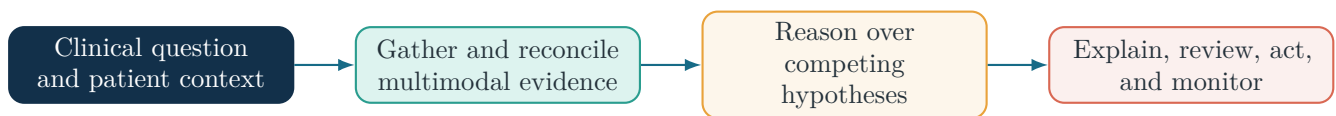


Figure 1: The system's clinical reasoning contract.

2 Cognitive Computing Architecture

2.1 End-to-end architecture

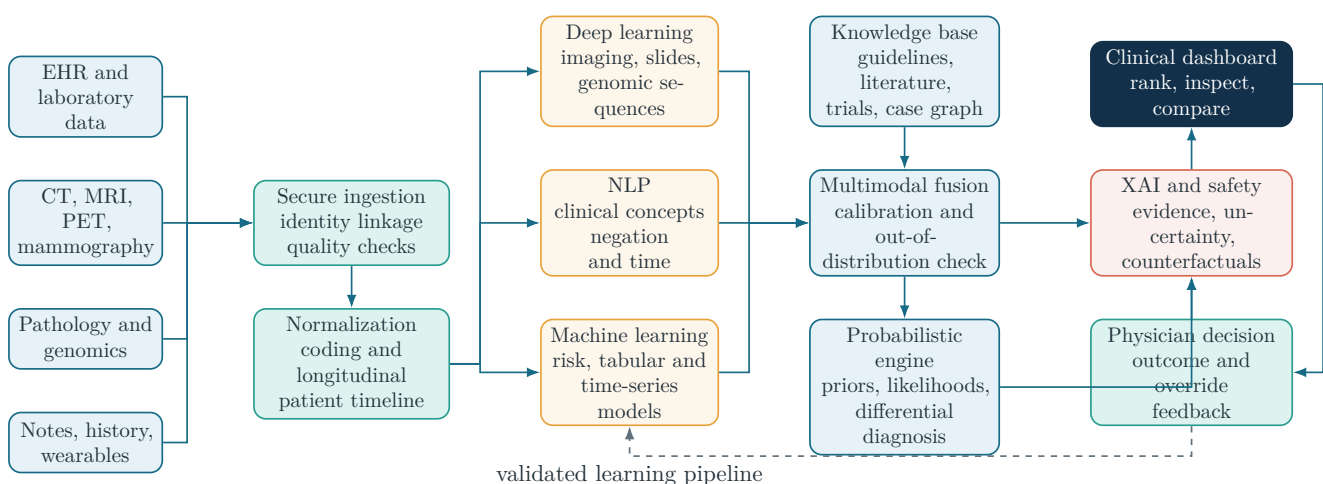


Figure 2: Reference architecture. Solid arrows represent case-time inference; the dashed arrow represents governed offline improvement.

2.2 Roles of the required cognitive components

Component	Primary role	Typical output for this case
NLP	Convert notes, pathology reports, and literature into structured, contextual clinical concepts; detect negation and temporality.	“No nodal involvement,” chronic cough onset, prior TB exposure, lesion descriptors, guideline passages.
Machine learning	Model structured clinical and laboratory features; stratify risk; estimate recurrence or response.	Tabular risk scores based on age, smoking, biomarkers, comorbidities, and trends.
Deep learning	Analyze images, whole-slide pathology, genomics, and other high-dimensional signals.	Lesion segmentation, margin irregularity, PET uptake pattern, tissue morphology.
Knowledge base	Ground reasoning in curated guidelines, staging criteria, drug knowledge, trials, and disease relationships.	Applicable staging rule, differential criteria, guideline version, evidence provenance.
Probabilistic reasoning	Combine priors and uncertain evidence; update a differential as evidence changes.	Calibrated probabilities for malignancy, TB, and benign nodule plus uncertainty.
Explainable AI	Make model behavior inspectable at patient and model level.	Supporting/opposing evidence, salient image regions, similar cases, counterfactuals.
Visualization	Present complexity as a decision workflow rather than a raw model score.	Ranked alternatives, timeline, trends, image overlays, next-test recommendations.

2.3 Multimodal evidence fusion

Safety boundary. If required evidence is missing, image quality is inadequate, or the patient lies outside validated populations, the system must lower confidence, explain the limitation, and allow abstention or escalation.

3 Kahneman’s System 1 and System 2

Kahneman’s distinction can be implemented as two cooperating computational modes. It is an analogy for workflow design, not a claim that a model literally thinks like a person.

3.1 Illustrated collaboration in the disputed lung case

- Step 1: Rapid screening.** Image and text models detect an irregular pulmonary lesion, weight loss, cough, and inflammatory evidence.
- Step 2: Initial hypotheses.** System 1 proposes lung cancer, tuberculosis, and benign pulmonary nodule.
- Step 3: Deliberate challenge.** System 2 asks whether infection can mimic the radiologic pattern, checks TB exposure, staging criteria, PET findings, pathology, and data quality.
- Step 4: Uncertainty-aware update.** Probabilities are recalibrated and contradictory evidence remains visible.
- Step 5: Information-seeking action.** The system recommends biopsy and molecular TB testing because these results are expected to separate the leading hypotheses.

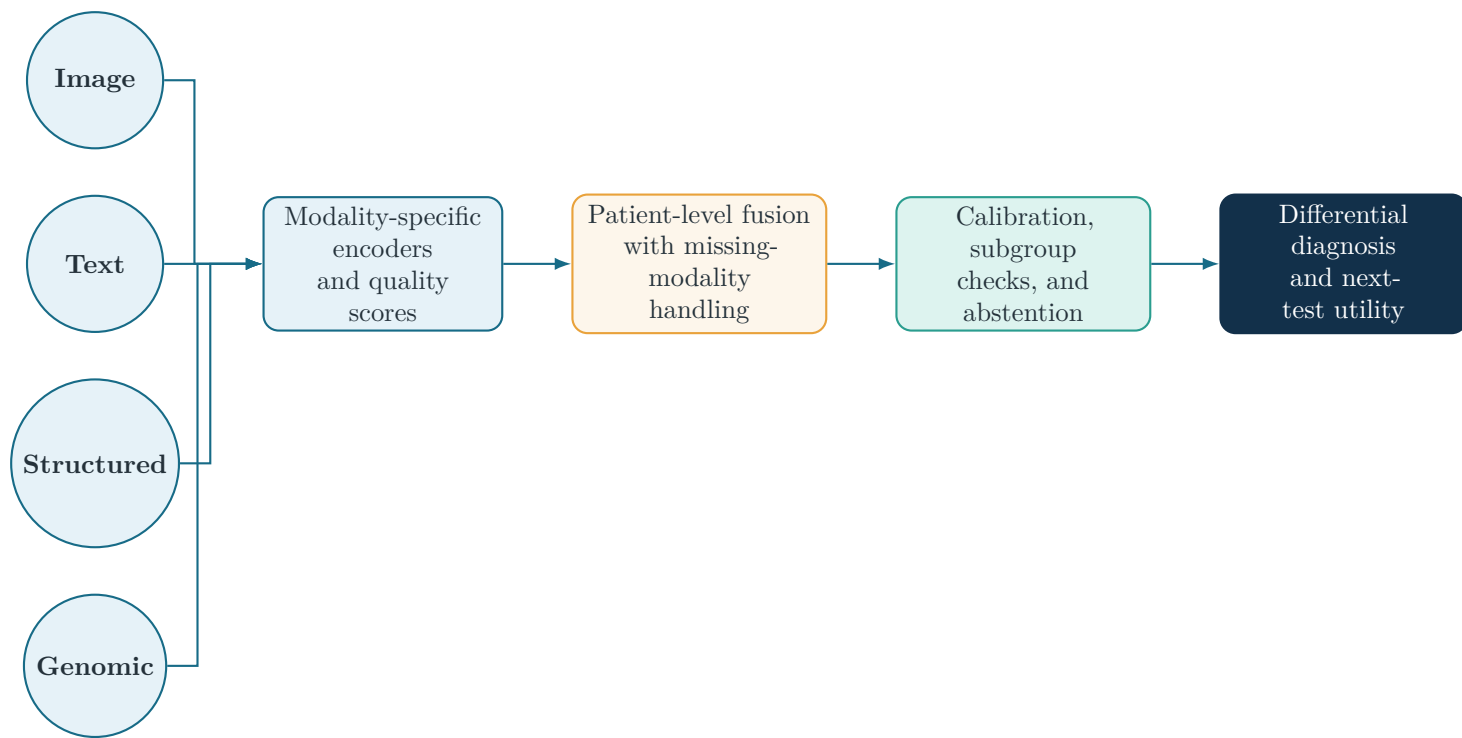


Figure 3: Fusion must retain modality quality and missingness, rather than pretending all inputs are equally reliable.

Step 6: Human resolution. The oncology team reviews evidence, patient preferences, test risks, and timing before deciding.

4 Rule-Based Expert System vs. Cognitive System

Dimension	Traditional rule-based expert system	Proposed cognitive system
Representation	Fixed IF–THEN rules	Learned models, knowledge graph, probabilistic dependencies
Inputs	Mostly manually structured fields	Structured, semi-structured, unstructured, images, genomics, time series
Uncertainty	Thresholds and deterministic conclusions	Calibrated distributions, confidence intervals, abstention
Adaptation	Experts manually revise rules	Controlled retraining using validated cases and feedback
Contradictions	Rule conflicts are brittle to resolve	Weights conflicting evidence and preserves alternatives
Imaging and text	Limited unless manually encoded	Native NLP and deep-learning analysis
Output	Often one categorical conclusion	Ranked differential, evidence, uncertainty, and next-test utility
Explanation	Trace of fired rules	Local evidence, visual attribution, provenance, counterfactuals, model card
Governance	Rule version and author	Data, model, knowledge, calibration, feedback, and override audit trail

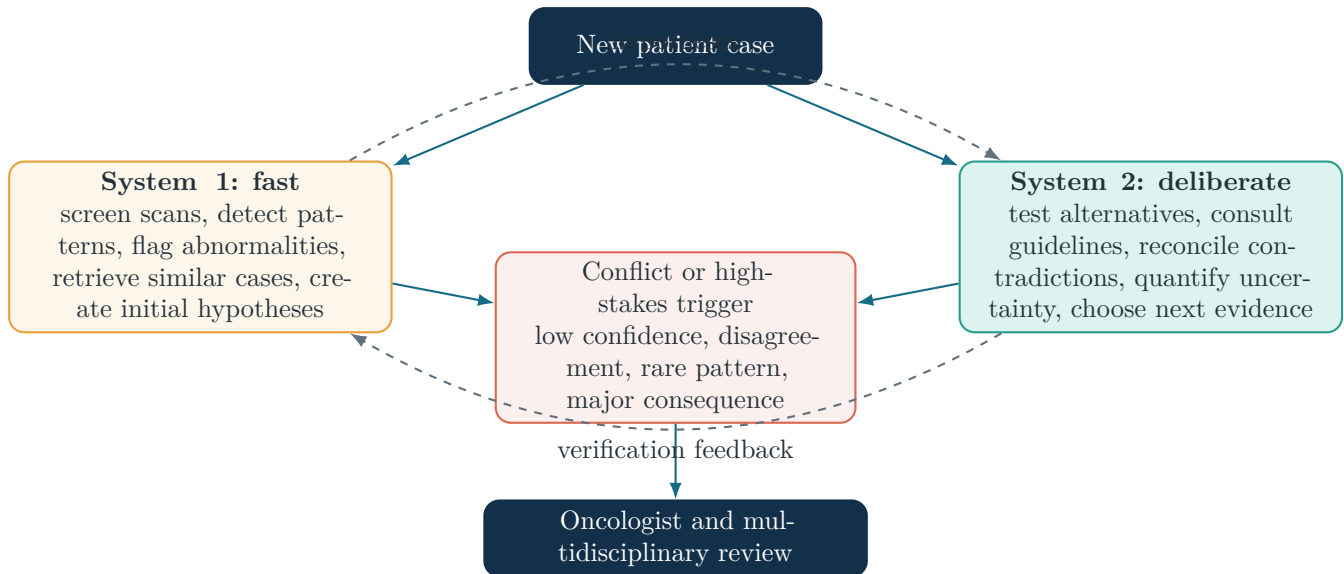


Figure 4: Fast recognition proposes; deliberate reasoning challenges; the clinician decides.

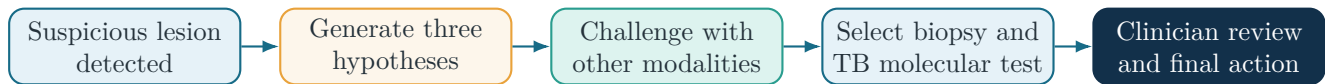


Figure 5: A practical System 1/System 2 sequence for the case.

4.1 Why probabilistic reasoning is appropriate

Symptoms overlap, test sensitivity and specificity are imperfect, biopsies can be non-representative, observations may be missing, and disease prevalence varies by population. Bayesian updating represents this uncertainty explicitly:

$$P(D_i | E) = \frac{P(E | D_i) P(D_i)}{\sum_j P(E | D_j) P(D_j)}.$$

Here, D_i is a candidate diagnosis and E is the accumulated evidence. A prior is not a fixed verdict; it is revised as evidence arrives.

Illustrative, not clinical probabilities: an interface might display lung cancer 62%, tuberculosis 27%, and benign nodule 11%. These numbers are useful only if calibrated on a relevant population and accompanied by uncertainty, provenance, and model limitations.

5 Five Explainable AI Features

5.1 1. Patient-specific evidence ledger

For every candidate diagnosis, show features that raise probability, features that lower it, missing evidence, and evidence quality. A doctor can distinguish “the model saw this” from “this is generally associated with the disease.”

5.2 2. Visual explanations for images

Overlay lesion segmentation, measurements, comparison with prior scans, and carefully validated saliency. The interface must warn that a heat map shows model attention, not causal proof or pathology confirmation.

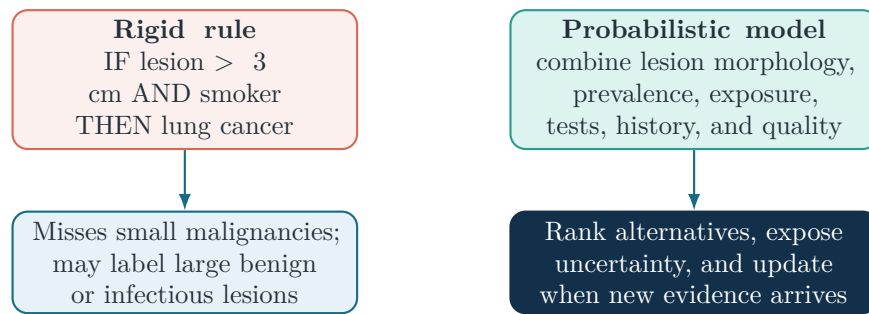


Figure 6: Deterministic thresholding versus evidence-weighted reasoning.

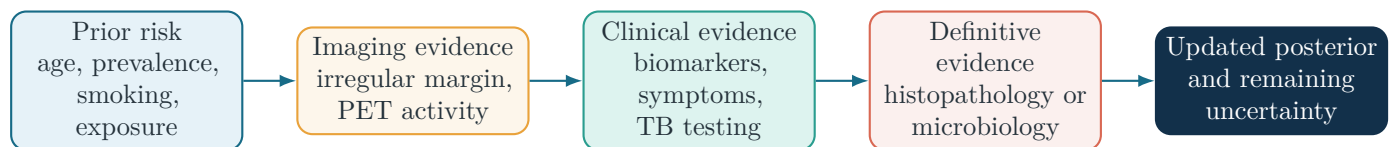


Figure 7: Sequential Bayesian updating as new evidence becomes available.

5.3 3. Ranked differential with calibrated uncertainty

Display multiple alternatives, credible or confidence intervals, calibration status, out-of-distribution warnings, and an abstention state. Include the next test expected to reduce uncertainty.

5.4 4. Guideline, literature, data, and model provenance

Link each recommendation to the exact guideline version, publication date, evidence passage, model version, data timestamp, and unavailable inputs. This forms a reproducible audit trail.

5.5 5. Counterfactuals, override, and accountability

Explain how plausible evidence changes would alter the ranking; allow doctors to override, record a reason, request multidisciplinary review, report bias, and attach the confirmed outcome. Preserve both the original output and the final human decision.

6 XAI Clinical Dashboard Concept

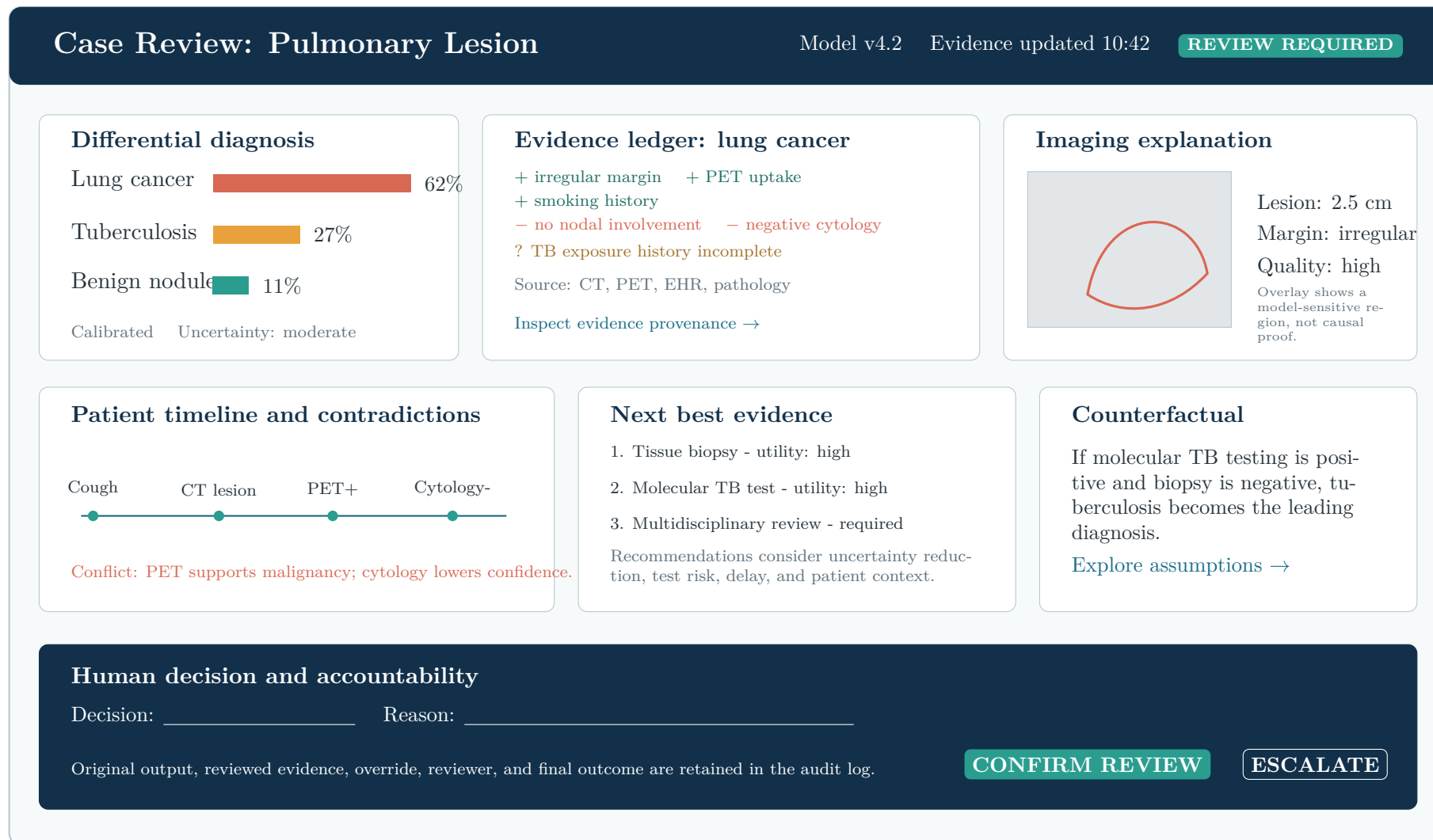


Figure 8: Conceptual dashboard showing the five XAI features as a single clinical workflow. Values are illustrative only.

7 Continuous Learning, Governance, and Accountability

Physician feedback is valuable only when linked to a reliable outcome and evaluated before deployment. Production models should be immutable during a case. New versions pass through offline curation, bias testing, clinical validation, approval, staged rollout, and monitoring.



Figure 9: Governed learning loop: feedback informs the next validated version, never an uncontrolled live mutation.

7.1 Risk controls

Concern	Design control	Evidence of accountability
Bias	Subgroup performance, calibration, data-representation checks, fairness review, external validation	Versioned evaluation report and deployment gate
Transparency	Patient-level evidence, model card, guideline provenance, known limitations	Reproducible explanation bundle saved with case
False positive	Confirmation workflow, threshold tuning, uncertainty display, test-risk awareness	Sensitivity/specificity by use case and error review
False negative	High-sensitivity screening where appropriate, escalation triggers, abstention, post-market monitoring	Missed-case audit and safety incident process
Ethical responsibility	Meaningful human control, informed workflow, privacy, access controls, patient-centered tradeoffs	Named clinical reviewer and decision rationale
Legal accountability	Intended-use definition, audit logs, validation records, change control, retention policy	Trace from data and model version to recommendation and final action

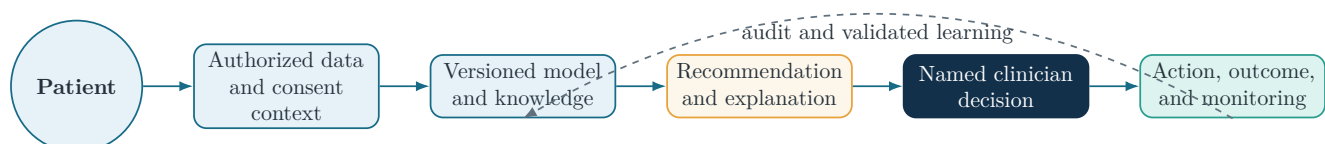


Figure 10: Accountability chain from patient context to monitored outcome.

8 Recommended Operating Workflow

- 1 Verify identity, permissions, data freshness, modality quality, and missing inputs.
- 2 Run modality-specific models and preserve their independent outputs.
- 3 Fuse evidence, retrieve applicable knowledge, and compute a calibrated differential.
- 4 Trigger deliberate review for disagreement, low confidence, rare cases, distribution shift, or high-consequence decisions.
- 5 Present evidence, uncertainty, provenance, counterfactuals, and next-test utility.
- 6 Require clinician review; record acceptance, modification, override, or escalation.
- 7 Link confirmed outcomes to the case and monitor errors, drift, and subgroup performance.
- 8 Retrain only through the governed validation and approval pipeline.

9 Conclusion

The proposed cognitive system combines NLP, machine learning, deep learning, a curated knowledge base, probabilistic reasoning, XAI, and visualization into one human-governed clinical workflow. System 1 supplies rapid pattern recognition and hypothesis generation; System 2 challenges those hypotheses, integrates contradictory evidence, and seeks the most informative next step.

Probabilistic reasoning is better suited than rigid rules because cancer diagnosis is inherently uncertain and multimodal. Trust comes not from a persuasive score, but from calibrated performance, visible limitations, traceable evidence, meaningful physician control, and a complete accountability chain.

AI proposes and explains. Clinicians interrogate and decide.
Governance makes both accountable.

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